

SMPS

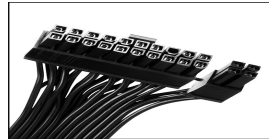
Switched mode power supply

- Another name for SMPS is **PSU-Power supply unite**
- SMPS is used mainly to convert AC into variable DC vlotage of 12v ,5v and 3.3v
- It will transfer not only AC to DC it will transfer DC to DC

Types of pins in smps

- **Main power connector**
 - Version 1--- 20 pins
 - Version 2--- 24 pins
- **CPU 4-pin & 8-pin connector**
- **Molex & SATA power connector**
- **PCI 6 pin & 8 pin power connector**

Main power connector



It is a main power connector it supplies power to your mother board and other components like **RAM** , **low end graphics card**

Earlier motherboard have 20-pins sokets only but now a days motherboards comes with 24-pins power connector

So all the SMPS comes with 24 pins detachable connector (20+4)

24-pin connector

- Orange 3.3v
- Red 5v
- Violet 5v
- Yellow 12v
- Blue -12v
- Gray (power boot)
- Black (ground)
- Green (power on)

CPU 4 pin power connector



This is a 12 volt power connector that supply power to your cpu.

Some motherboare have 4pin 12v CPU socket and some have 8pin 12v socket.

Therefore most of the power connectors are comes with 8pin (4+4) detachable connector which can be split into two 4pin connector.

CPU 4pin

- Yellow 12v
- Yellow 12v
- Black (G)
- Black (G)

Molex & SATA power connector



Molex 4pin power connector is used to provide power supply for hard drive and IDE drive

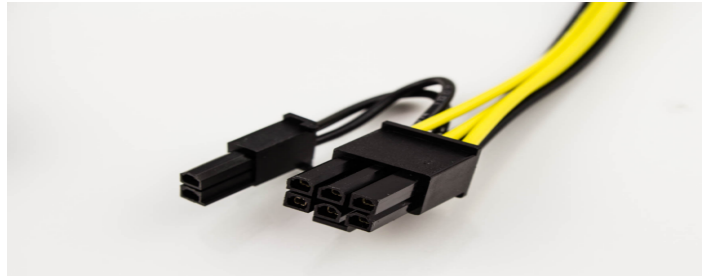
In older device we use molex 4pin connector but now a days the molex is not using

We using SATA power connector

Molex & SATA power connector

- Yellow 12v
- Black (G)
- Black (G)
- Red 5v

PCI 6pin & PCI 8pin



Peripheral component interconnect there are two ranges one is 6pin and another one is 8pin

This connector is used to provide power supply for mid range graphics card

For high range graphics card we need 8 pin

PCI 6 pin & PCI 8 pin

- Yellow 12v
- Black (G)
- Black (G)
- Red 5v

Types of smps

- Non-modular
- Semi-modular
- Fully-modular

Non-modular



Semi-modular



Fully-modular

